



Introduction to Computer

The objective of this course is to acquaint the students with the basics of computers system, its components, data representation inside computer and to get them familiar with various important features of procedure-oriented programming language i.e. C.

This Course guides the students to read, write and modify C programs and to implement basic projects

Course Objectives

Introduction to Computer



Computer is an integral part of our day to day life.

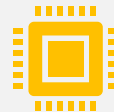
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Meaning of the computer is a device that can calculate.



But the modern computer can do a lot more then calculation.



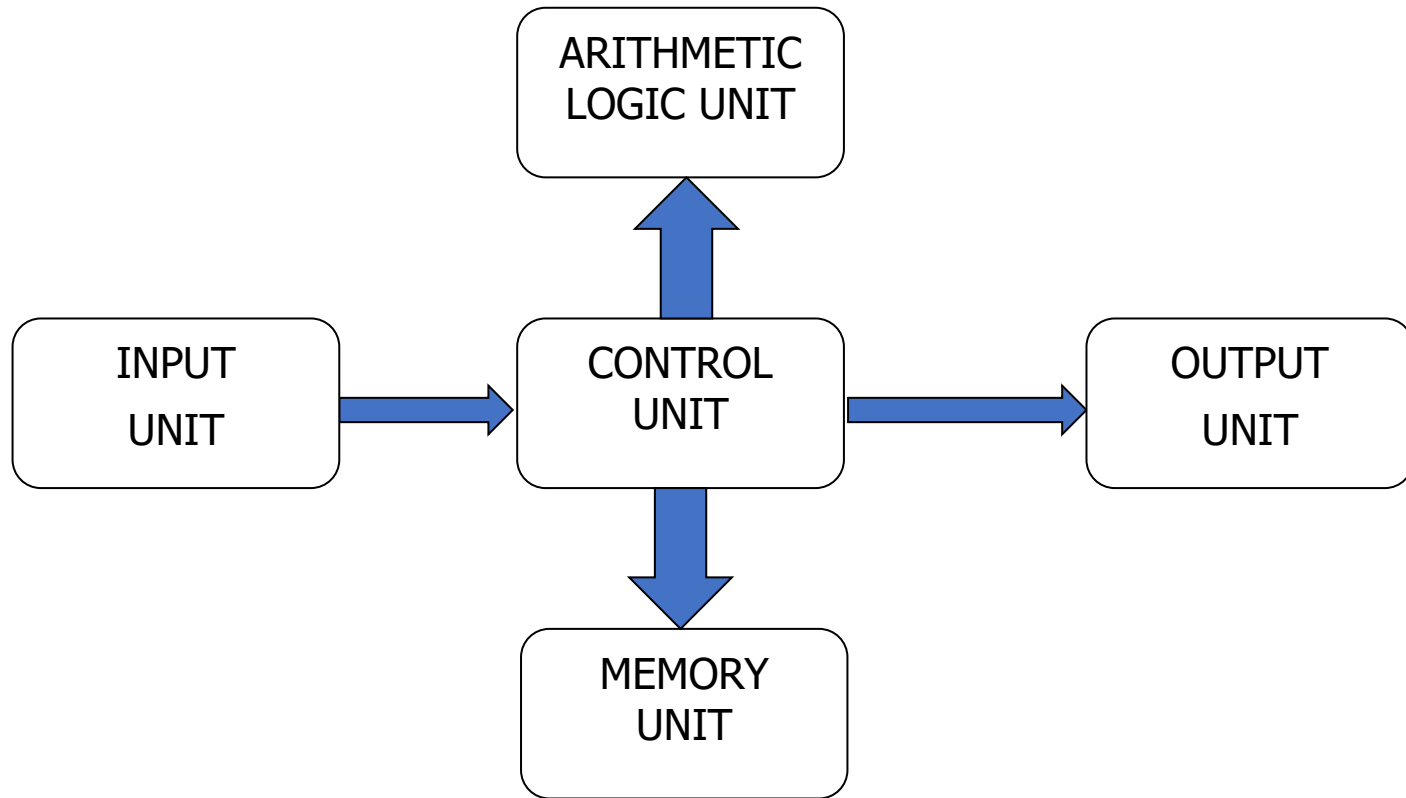
Computer is an electronic device that receive input, stores or processes the input as per user instruction and provides output in desired format.

Introduction to Computer



- Computer input is called data.
- Output obtained after processing data based on user's instructions is called information.
- Raw fact and figures which can be processed using arithmetic and logical operations to obtain information are called data.
- Processes that can be applied to data can be two type
 - Arithmetic Operation
 - Logical operation

Block Diagram of Computer



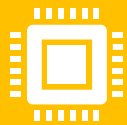
Component of Computer



INPUT UNIT: Devices used to input data and instructions.

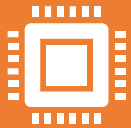


OUTPUT UNIT: Devices used to provide information to the user in desired format.



CONTROL UNIT: All devices or parts of computer interact through the control unit.

Component of Computer



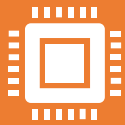
ARITHMETIC LOGIC UNIT: All arithmetic operation and logical operations take place through this unit.



MEMORY UNIT: All input data ,information and data interim to the processes are store in the memory. Memory of two type-

primary memory
secondary memory

Component of Computer



Control unit, arithmetic logic unit and memory are together called the central processing unit or CPU.



Computer devices that we can see, and touch are the hardware components of a computer.



Set of instruction or programs that make the computer function using these hardware point are called software.

Characteristic of Computer



SPEED: Typically a computer can carry out 3-4 million instruction per second.



ACCURACY: Computers exhibits a very high degree of accuracy. Errors that may occur are usually due to human errors.



RELIABILITY: Computer can carry out same type of work repeatedly without throwing up errors due to tiredness or boredom.

Characteristic of Computer



VERSATILITY: Computer can carry out a wide range of work.



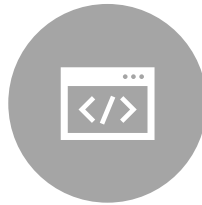
STORAGE CAPACITY: Computer can store a very large amount of data at a fraction of cost of traditional storage of files.

Advantages and Disadvantages of using Computer

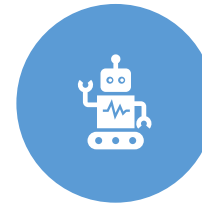


ADVANTAGE:

COMPUTER CAN DO THE SAME TASK REPETITIVELY WITH SAME ACCURACY.



COMPUTER DO NOT GET TIRED AND BORED



COMPUTER CAN TAKE UP ROUTINE TASKS WHILE RELEASING HUMAN RESOURCE FOE MORE INTELLIGENT FUNCTION.



DISADVANTAGES :

COMPUTER HAVE NO INTELLIGENCE



REGULAR ELECTRIC POWER SUPPLY IS REQUIRED.



TRAINED PEOPLE REQUIRED TO WORK ON COMPUTERS.

- First Generation: (1942-1954)
 - Vacuum tube technology
 - Unreliable
 - Very costly
 - Generate lot of heat
 - Huge size
 - Need AC
 - Supported machine language only
 - Consumed lot of electricity
 - Example:
ENIAC, EDVAC, IBM-701, IBM-650

- **History of Computers**

- Second Generation: (1952-1964)
 - Use of transistors
 - Reliable as compared to first generation
 - Smaller size as compare to first generation
 - Generate less heat as compare to first generation
 - Consumed less electricity as compare to first generation
 - Faster as compare to first generation
 - Still very costly
 - Need AC
 - Supported machine language and assembly language Example: IBM-1620, IBM-7094

- **History of Computers**

- Third Generation:
(1964-1972)
 - IC used
 - More Reliable
 - Smaller size
 - Generate less heat
 - Consumed less electricity
 - Faster
 - Still costly
 - Need AC
 - Supported high level language
 - Example: IBM-360 series ,Honeywell-600

- **History of Computers**

- Forth Generation: (1972-1990)
 - VLSI
 - More Reliable
 - Smaller size
 - Use of PC's
 - Pipeline processing
 - Concept of internet and network was introduced
 - Very cheap
 - No AC needed
 - Computer become easily available
 - Example: DEC-10, STAR-1000, CARY (super computer)

- **History of Computers**

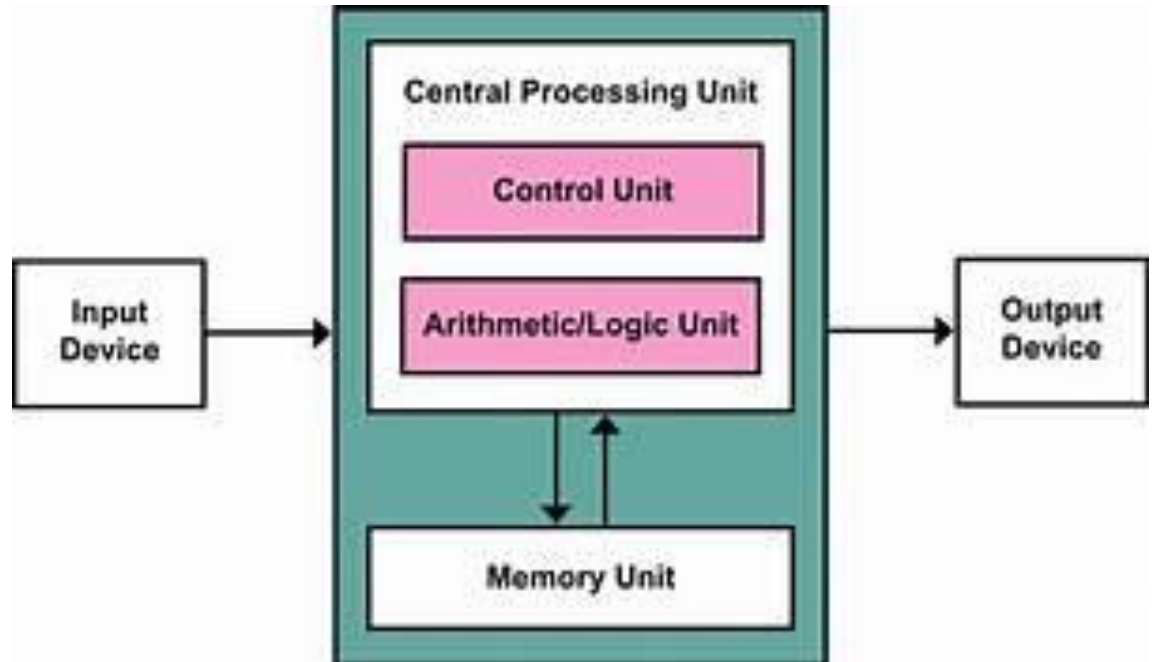
- Fifth Generation:
(1990 – till date)
 - ULSI
 - C,C++, JAVA, .NET
 - Development of AI
 - Development of NLP
 - Advancement in parallel processing
 - More user-friendly interface with multimedia feature
 - Example: Desktop, laptop, notebook.

- **History of Computers**

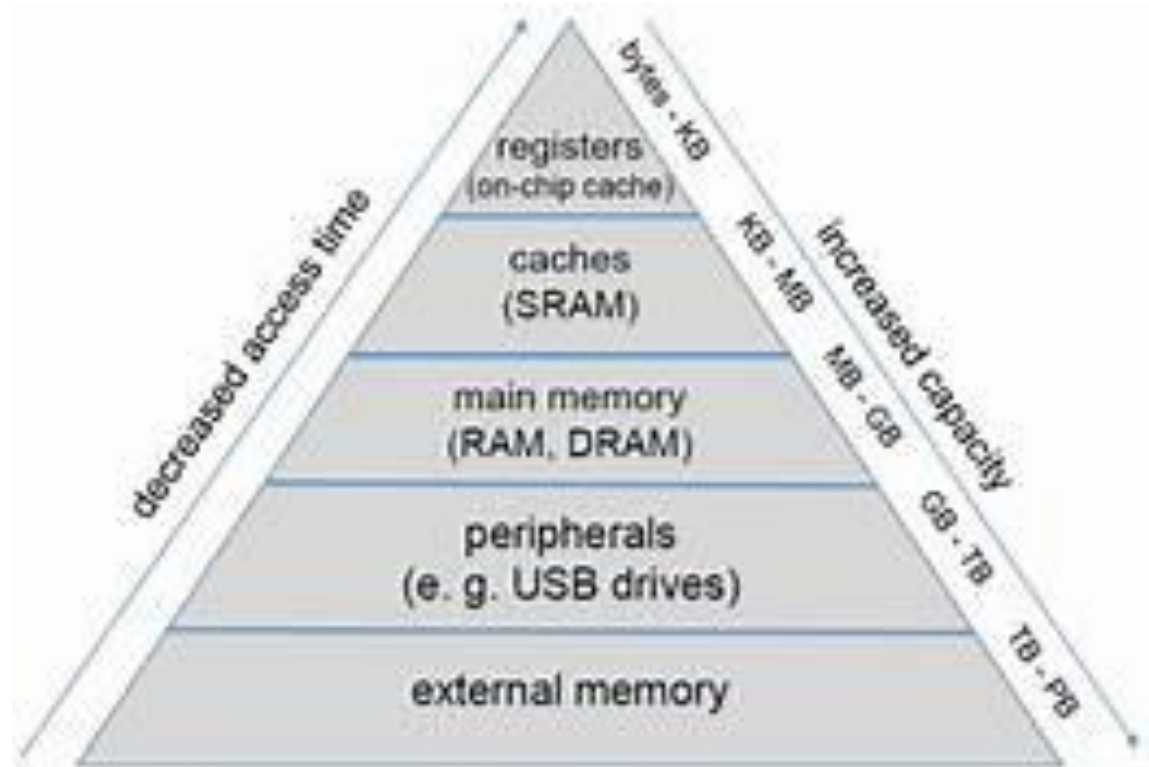
- Almost all computer design are based on the concepts developed by John Van Neuman. Such design are referred to as Von Neuman architecture.
- The 3 main Concept:
 - Data and instructions are stored in a single read write memory.
 - The contents of this memory are addressable by location irrespective of the type of data or instruction present there.
 - Execution occurs in the sequential manner from one instruction to the next unless changed explicitly.

- **Von-Neuman Architecture**

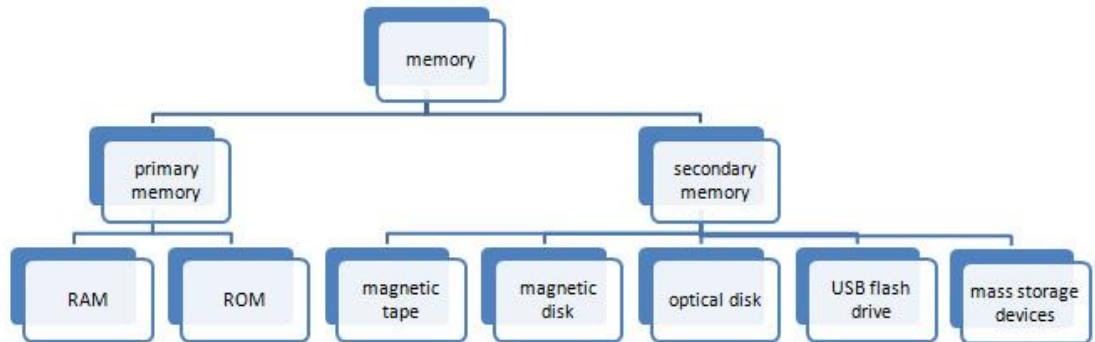
Von-Neuman Architecture



Memory Hierarchy



Types of Memory



- **Capacity :** It is the global volume of information the memory can store. As we move from top to bottom in the Hierarchy, the capacity increases.
- **Access Time:** It is the time interval between the read/write request and the availability of the data. As we move from top to bottom in the Hierarchy, the access time increases.
- **Performance:** Earlier when the computer system was designed without Memory Hierarchy design, the speed gap increases between the CPU registers and Main Memory due to large difference in access time. This results in lower performance of the system and thus, enhancement was required.
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• Characteristics of Memory

- This enhancement was made in the form of Memory Hierarchy Design because of which the performance of the system increases. One of the most significant ways to increase system performance is minimizing how far down the memory hierarchy one has to go to manipulate data.
- **Cost per Bit:** As we move from bottom to top in the Hierarchy, the cost per bit increases i.e. Internal Memory is costlier than External Memory.

- **Characteristics of Memory**

- Input/output devices are capable of sending data (**output**) to a computer and receiving data from a computer (**input**).

Example:

- **CD-R/RW, DVD, and Blu-ray drive**
- **Digital camera**
- **Hard drives**
- **Modem**
- **Network adapter**
- **SD Card**
- **Touch screen**
- **USB thumb drives**
- **Printer**

- **I/O Devices**

- **System Software:** Software which require H/W parts of the computer and application software to run.
- Acts as interface between hardware and user application
- Interface is needed because hardware devices or machine and human speak in different languages.
- English is the pre-dominating language of interacting with computers.
- System software convert all human instructions into machine understandable instruction.
- **Type of System Software:**
 - **Operating system**
 - **Language Processor**
 - **Device Drivers**

- **Type of Software**

- **Operating System:** System software responsible for functioning of all hardware parts and their inter operativity to carry out task successfully.
- First software to be loaded in computer memory
- Manages a computer's basic functions like storing data in the memory , retrieving files from the storage device, schedule tasks based on priority.
- **Language Processor:**
Convert all user instructions into machine understandable language.

- **Operating System**

- In terms of human machine interaction, languages are of three types:
- **Machine Level Language:**
A string of 0s and 1s that the machine can understand.
- **Assembly Level Language:**
Define Mnemonics
- **High Level Language :** User English like statement and it is completely independent of machine
- Program written in high level programming language like JAVA, C++, etc. is called source code
- Set of instructions in machine readable form is called object code or machine code.
- System software that converts source code to object code is called language processor.

- **Language Processor**

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- **Language Processor**

- Three types of language interpreters:
- **Assembler:** Convert assembly language program into machine level language.
- **Interpreter:** Convert high level program into machine level program but line by line
- **Compiler:** Convert high level program into machine level program at one go rather than line by line.

- **Language Processor**

- System software that controls and monitors the functioning of a specific device on computer.
- Each device that needs to be attached externally to the system has a specific driver associated with it.
- When you attach a new device , you need to install its driver so that the OS knows how it need to be managed

- **Device Drivers**

- A software that performed a single task.
- Here are some commonly used application software:
 - Word Processing
 - Spread Sheet
 - Presentation
 - Database Management
 - Multimedia tools

- **Application Software**

Books

- ANSI 89 – American National Standards Institute, American National Standard for Information Systems Programming Language C, 1989.
- Kernighan 78 – B. W. Kernighan and D. M. Ritchie, The C Programming Language, Prentice-Hall: Englewood Cliffs, NJ, 1978. Second edition, 1988.
- Thinking 90 – C* Programming Guide, Thinking Machines Corp. Cambridge Mass., 1990.
- Programming in ANSI C, E. Balagurusamy. McGrawHill
- Let Us C, Yashvant Kanetkar, BPB Publications
- Programming in C, Reema Thareja, Oxford University Press

Thank You

